

Technical Document #213: Retrieval Augmented Generation - Comprehensive Overview

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Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Hallucination detection identifies when LLMs generate factually incorrect information.